Project 1: Eagle Delivery — Technical Report

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1. Introduction & Problem Formulation

The Eagle Delivery project models an autonomous campus courier robot tasked with delivering packages between building entrances across Eastern Michigan University. The system is evaluated under two distinct paradigms: informed graph search, where the robot leverages complete map knowledge to compute optimal paths, and tabular Q-learning, where the agent learns effective navigation policies purely through trial-and-error environmental interaction without prior map topography.

Formally, the campus environment is structured as a discrete grid-world defined as a 6-tuple `(S, s0, A, T, G, c)`:

- **State Space (S):** Every traversable coordinate position `(row, col)` corresponding to walkways (`.`) and building entrances (`S, W, L, D, H, P`).
- **Initial State (s_0):** The designated coordinate of the starting building entrance for a given delivery task.
- **Actions (A):** Four discrete movement operations: `UP`, `DOWN`, `LEFT`, and `RIGHT`. Wall collisions or boundary attempts result in position persistence.
- **Transition Model (T):** Deterministic coordinate shifts by standard directional offsets unless blocked by a wall cell (#).
- **Goal Test (G):** An episode terminates successfully when the agent's current position matches the target building's coordinate.
- **Path Cost (c):** Every attempted action incurs a unit cost of `1`. Because edge costs are uniform and positive, Breadth-First Search (BFS) is guaranteed to find the shortest path.

2. Manhattan Distance Admissibility Proof

For informed graph search, A* employs the Manhattan distance heuristic, defined as $h(s) = |r_s - r_g| + |c_s - c_g|$. To guarantee that A* returns optimal paths, $h(s)$ must be admissible, meaning it never overestimates the true remaining cost to reach the goal: $h(s) \leq h^*(s)$ for all states s.

Proof of Admissibility: Consider an unconstrained grid where all obstacle walls are removed. In this relaxed 4-connected grid, the shortest path between any two cells $s = (r_s, c_s)$ and $g = (r_g, c_g)$ requires exactly $|r_s - r_g|$ vertical steps and $|c_s - c_g|$ horizontal steps. Thus, the true optimal cost in the relaxed graph is exactly $h^*(s) = |r_s - r_g| + |c_s - c_g|$. When physical obstacles (walls #) are introduced, the true shortest path distance $h^*(s)$ can only increase around barriers. Therefore, $h(s) \leq h^*(s)$ holds universally, proving admissibility. Furthermore, the heuristic satisfies consistency: $|h(s) - h(s')| \leq 1$ for any adjacent action transition.

3. Part A: Search Comparisons & Results

Both BFS and A* were executed across six benchmark scenarios combining the baseline campus map (`emu_campus.txt`) and the obstructed detour map (`emu_detour.txt`). Table 1 summarizes the route lengths and expanded node counts.

Map Environment	Delivery Task	BFS Length	A* Length	BFS Exp.	A* Exp.
Baseline (emu_campus.txt)	Student Center → Walton	18	18	67	22
Baseline (emu_campus.txt)	Halle Library → Downing	22	22	79	62
Baseline (emu_campus.txt)	Pray-Harrold → Putnam	16	16	41	16
Detour (emu_detour.txt)	Student Center → Walton	26	26	70	51
Detour (emu_detour.txt)	Halle Library → Downing	22	22	65	60
Detour (emu_detour.txt)	Pray-Harrold → Putnam	16	16	41	16

Discussion: Both algorithms consistently find paths of identical length, confirming A* optimality. A* significantly reduces node expansions (e.g., expanding 22 nodes vs. 67 for S → W baseline) by directing expansion toward the goal.

4. Part B: Q-Learning Formulation & Baselines

To handle environments without prior maps, tabular Q-learning updates action-value estimates $Q(s,a)$ via temporal differences: $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s,a)]$. Hyperparameters: learning rate $\alpha = 0.3$, discount factor $\gamma = 0.95$, exploration rate $\varepsilon = 0.20$ constant, trained for 2,000 episodes across seeds 0, 1, and 2. Evaluated over 100 episodes with learning frozen ($\varepsilon = 0.0$). All three baseline tasks (S → W, L → D, H → P) successfully converged to a **1.0 mean success rate** across all seeds, achieving exact optimal mean returns of 22.0, 18.0, and 24.0 respectively.

5. Part C: Custom Campus Map Design & Epsilon Study

A custom map (`maps/my_campus.txt`, 10×15 grid featuring direct corridors, cyclic loops, and dead-end branches) was developed and validated. A* search successfully solved the custom route in 11 expanded nodes with an 11-step path length. Furthermore, controlled training experiments varying exploration rates ($\varepsilon \in \{0.05, 0.20, 0.50\}$) across seeds 0, 1, and 2 demonstrated that off-policy Q-learning robustly converges to the identical optimal 11-step greedy policy (success rate 1.0, mean return 29.0) regardless of exploration intensity. However, higher exploration rates ($\varepsilon = 0.50$) increased online dithering penalties during training.

6. Conclusion

This project successfully implemented and evaluated informed graph search and reinforcement learning agents for autonomous campus navigation. Empirical evaluations verify that A* substantially reduces computational overhead while preserving route optimality, and tabular Q-learning successfully masters environmental dynamics through direct interaction. All codebase unit tests (14/14) pass successfully, and experimental results are fully reproducible via the provided repository scripts.